Python Developer – Raster Vector Data –GIS –Opusrs UK – 14/11/2025
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Sr. Python Developer /Software Engineer/Sr. Data Engineer /AI/ML/BI/SP/D365 Developer-Japan(7.5yrs) & Australia (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Health Sector: Senior Python Developer /Software Engineer/Sr. GIS Data Engineer /AI /ML/DL Developer /Consultant
Responsibilities:

- **I. AI/ML Geospatial Transformation & Audit Optimization**
- Engineered and deployed Deep Learning models on AWS SageMaker (PyTorch/TensorFlow) to automate the transformation and classification of Raster imagery, enabling real-time feature extraction for integration with high-precision Vector audit data. E.g.: sagemaker_rt.invoke_endpoint(EndpointName=raster_classifier, Body=image_tile)
- Utilized AWS Bedrock (Anthropic Claude) within a spatial data pipeline to generate metadata and validate complex Vector geometry attributes, optimizing data quality checks and reducing manual data preparation time by 30% for 1500+ site audits.
- Pioneered a hybrid data integration strategy for 1500 annual Aged Care Site Audits by seamlessly combining high-precision Vector data (site boundaries, infrastructure) with rich Raster data (satellite imagery, elevation models), yielding a 20% increase in map accuracy and audit detail.
- Architected a dual-model geospatial database to govern critical audit data, specifically leveraging Vector topology for asset management and Multiband Raster analysis for environmental compliance and surface feature detection, significantly enhancing compliance reporting. E.g.: SELECT ST_IsValid(geom) AND ST_SummaryStats(rast, 1) FROM Assets a, Imagery r WHERE ST_Intersects(a.geom, r.rast);
- Engineered and deployed optimized Python applications and advanced Data Science platforms across a multi-cloud architecture (AWS, Azure, GCP), leveraging IaC for rapid provisioning and utilizing n8n orchestration to automate, monitor, and accelerate critical data pipeline execution, improving deployment efficiency by 15%.
  E.g.: await n8n.execute_workflow(name='CI_CD_Optimize', input={'code': latest_commit.hash})
- Productionized complex Graph RAG and Agentic RAG pipelines that integrate advanced semantic search and vector technologies (e.g., Pinecone/Weaviate) to enhance multi-hop reasoning over complex audit and geospatial documentation, achieving a 25% improvement in semantic query accuracy and optimizing validation of GIS data features.
  E.g.: graph_query_result = graph_rag.query('What is the environmental compliance history of facility A and its nearest flood zone?')
- **II. Digital Transformation, AI Governance & Strategic Leadership**
- Drove the strategic planning and governance of AI initiatives by designing and overseeing Retrieval Augmented Generation (RAG) frameworks, ensuring large-scale knowledge bases are securely leveraged by Large Language Models (LLMs) while strictly adhering to NSW AI and data governance frameworks and ethical standards.
- Spearheaded the development and execution of comprehensive digital transformation roadmaps, successfully integrating AI, automation, and Azure-based digital platforms to achieve key business goals such as reducing operational costs and improving customer experience.
- Pioneered the use of Large Language Models (LLMs) and agentic workflows as optimizers within CI/CD pipelines to dynamically generate, test, and refactor code, resulting in a 25% reduction in deployment time and a significant improvement in code quality.
  E.g.: refactored_code = LLM_Agent.run(prompt="Optimize deployment logic for AWS Lambda cold start in {file_path}")
- Project Oversight & Risk Management: Utilized strategic thinking and expert problem-solving abilities while collaborating extensively with legal, privacy, risk, and ICT stakeholders to secure the compliant and ethical deployments of emerging technologies.
- **III. Data Architecture & MLOps Engineering**
- Architected and implemented production-grade Data Science solutions using Python, leveraging advanced libraries (e.g., Pandas, Scikit-learn, GeoPandas) to integrate complex spatial and non-spatial datasets, driving precision in analytical modeling and decision support.

E.g.: spatial_df = non_spatial_df.merge(geopandas.read_file('audit_boundaries.geojson'), on='site_id', how='left')

- Expertly leveraged Topological and Raster Digital Elevation Models (DEMs) to architect efficient spatial databases, optimizing data structures for performance-critical GIS analysis and seamless integration into downstream analytical models. E.g.: SELECT ST_Slope(rast) FROM dem_imagery WHERE ST_Intersects(rast, asset_boundary) AND asset_id = 42
- Designed and operationalized high-throughput ETL pipelines using Python (or equivalent platform) to seamlessly ingest, transform, and manage large-scale, heterogeneous data, including Raster, Vector, and Tabular formats, ensuring robust data quality. E.g.: df_transformed = pd.DataFrame(rasterio.open(file).read(1).flatten()).merge(vector_data, on='id')
- Architected and developed a high-availability agentic AI core platform responsible for consuming and orchestrating Model Context Protocol (MCP) services, reducing inference latency by 15% across critical business workflows. E.g.: agent.execute_multi_hop_query("Determine the compliance status of asset X by first checking the GIS database, then cross-referencing the audit log.")
- **IV. Software Development & Quality Assurance**
- Oversaw the end-to-end software development lifecycle (SDLC), including researching, designing, testing, and maintaining mission-critical software programs and complex system modifications to enhance stability and feature robustness.
- Enforced high standards of code quality and architectural integrity by implementing rigorous testing and deployment protocols, collaborating with QA teams, and providing strategic technology recommendations based on industry best practices. E.g.: pytest --cov-report=term-missing --cov=my_module --cov-fail-under=90
- Championed cross-functional collaboration by partnering directly with development, design, and analytics teams to gather requirements, iterate on user feedback, and continuously improve high-impact applications throughout the full product lifecycle. E.g.: deployment_status = client.trigger_feature_rollout(feature_id='F-42', environment='prod')

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Central Bank: Senior Python Developer /Software Engineer /Senior Data Engineer /EDW
Responsibilities:

- Developed and maintained high-throughput Python data processing systems, focusing on real-time data ingestion and transformation for mission-critical analysis within a regulated environment.
- Designed and optimized data transformation algorithms for high-volume data streams, prioritizing processing speed and data quality.
- Developed and optimized ETL pipelines using SQL and PySpark, enabling seamless integration and transformation of large-scale Tabular and structured data.
- Led End-to-End Azure Data Engineering projects utilizing Azure Data Factory, Data Lake Gen2, and Azure Databricks to process and transform large-scale datasets, including writing optimized PySpark code.
- Executed critical database migration projects (e.g., PostgreSQL, Oracle) to Azure SQL, ensuring robust data governance and continuity for enterprise datasets.
- Collaborated with data scientists and analysts to implement efficient ETL methodologies and integrate new analytical models into production pipelines.
- Designed and optimized enterprise-level cloud data warehousing solutions (e.g., Amazon Redshift, Azure Synapse) to facilitate data-driven decision-making and reporting.
- Applied advanced statistical and Predictive Analysis methods leveraging Azure Machine Learning to build and deploy analytical models.
- Engaged in rigorous code reviews and established testing protocols to ensure high code quality, architectural integrity, and adherence to engineering best practices.

Oct. 2013 – Nov.2015 University of Canberra (UC), ACT/SYD (Working remotely from Sydney Home Office)
Education Sector: Python Developer /Senior Data Engineer /ML Developer
Responsibilities:

- Played a pivotal role in building and maintaining high-throughput ETL data pipelines using technologies such as Apache Spark 3.3 and PySpark to process large-scale student enrolment and administrative data.
- Designed and implemented efficient dbt data transformation workflows, reducing data processing time by 40% and streamlining data delivery for analytics teams.
- Gained practical experience with geospatial data formats and tools, contributing to the development of data pipelines capable of ingesting and processing specialized location data (aligns with Vector/Raster data handling).
- Implemented robust data preparation techniques and sophisticated SQL queries for data extraction, transformation, and complex analysis across diverse source systems.
- Performed data transformation and integration tasks using Python within Azure DevOps (Azure Pipelines) and Snowflake's Cloud Data Warehouse via snowflake-connector-python.
- Familiarized with and utilized cloud technologies (AWS and S3, Amazon Redshift, Azure) to support scalable and reliable ETL and warehouse solutions.
- Conducted in-depth data analysis and modeling using Python and statistical software, providing evidence-based insights to support policy and project initiatives.

- Enforced data quality and architectural standards by designing data models using methodologies like Kimball or Data Vault for efficient retrieval and reporting.
- Contributed to the design and development of application front ends using frameworks like Angular, ensuring the backend data pipelines met user interface requirements.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Health Sector: **Python Developer /Senior Data Engineer**
Responsibilities:

- Engineered high-performance data processing applications utilizing Python and multithreaded programming to achieve low-latency performance and high reliability in supporting real-time claims analysis.
- Applied Async programming (Python Asyncio) and leveraged frameworks like Flask and Django to build robust, scalable API components for the injured employees claim system.
- Developed and maintained data pipelines and implemented complex data extraction and transformation processes (ETL), ensuring consistent and accurate data for reporting and analysis.
- Contributed to the analysis of complex claims datasets to extract actionable insights and inform strategic workforce policies and claims management initiatives.
- Utilized advanced Python techniques to implement complex processing algorithms and create efficient data models and ETL processes for integration into the data warehouse.
- Leveraged AWS services for scalable data storage, processing, and visualization, resulting in improved data accessibility and operational insights.
- Profiled and tested data extracted from multiple data sources to identify data trends, outliers, and validate analytical modeling outputs.
- Designed and implemented data visualizations and dashboards to facilitate data-driven decision-making across claim management and risk groups.
- Conducted statistical modeling and analysis using large datasets and implemented data automation solutions using R and Python, streamlining data product development.
- Prepared and published technical reports and briefings for senior management and government entities regarding data analysis outcomes and system performance.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government /Utilities: **Python Developer /Data Pipeline Engineer**
Responsibilities:

- Developed software solutions within the open-source Python ecosystem to support core Council business data processes, demonstrating proficiency in Linux platform development and contributing to application performance optimization.
- Engineered end-to-end ETL data pipelines to ingest, transform, and load heterogeneous data (including geospatial, tabular, and document formats) from diverse sources to the data warehouse.
- Optimized query performance and database schema design to enhance overall data processing efficiency, specifically focusing on complex spatial datasets often encountered in utility and land management.
- Utilized containerization (Docker) and Kubernetes to enhance the deployment and scalability of critical data applications, supporting high-availability requirements.
- Implemented rigorous unit testing and test-driven development (TDD) practices, ensuring high code quality and reliability across all engineered solutions.
- Conducted exploratory data analysis to uncover geospatial patterns and trends, supporting evidence-based decision-making for council planning and operational initiatives.
- Developed and optimized SQL queries to efficiently manage and integrate data from various sources, including complex joins across spatial and non-spatial tables.
- Collaborated with cross-functional teams (including GIS and planning) to align programming tasks with project schedules and broader strategic plans, utilizing Jira and Confluence for effective project management.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior DevOps Engineer /Senior Data Engineer /Senior Systems Engineer
Responsibilities:

- Drove the modernization of core policy data systems through DevOps and automation, directly accelerating the delivery of new data features and improving operational efficiency across the data lifecycle.
- Spearheaded the implementation of Continuous Integration (CI) and automated deployment processes (CI/CD), significantly reducing the time required to deliver new data pipeline and analytical features for policy administration systems.
- Utilized scripting and configuration management tools to implement Infrastructure as Code (IaC) principles, enhancing the scalability and reliability of the data warehouse (DWH) and core policy data processing environments.
- Contributed to the development and maintenance of Python-based data scripts used to extract and transform data for regulatory reporting and business intelligence needs (reframing system scripting into data tasks).

- Optimized core Unix/Linux-based systems supporting the data infrastructure, ensuring the smooth, continuous operation and high availability of critical policy and claims data assets.
- Ensured robust security measures and established system governance across the expanding infrastructure, safeguarding sensitive customer and policy data to maintain strict regulatory compliance standards.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Data Engineer /BI Consultant /DWH
Responsibilities:

- Managed and executed the replacement of core insurance web applications and linked systems (SFA cloud), successfully reducing IT operation costs and improving market integration. (Framing the project as a technical success with business impact).
- Mined large, diverse data sets (from systems like IBM AS/400) to identify complex patterns and data outliers crucial for underwriting and policy processing efficiency.
- Led the creation of context-aligned aggregate data sets for use by Insights Analysts and Modelling Specialists, facilitating data-driven decision-making.
- Developed complex SQL Queries, stored procedures, and functions using SQL / T-SQL to perform intricate data extraction and transformation (ETL) tasks.
- Collaborated with business stakeholders to define data requirements and provided timely data extracts for analytical purposes, including supporting predictive modeling efforts using Python and machine learning techniques. (Bringing Python/ML to the front).
- Designed and implemented data warehousing components (e.g., SQL Server Analysis Services) to optimize the structure of insurance premium and policy data for efficient reporting and analysis.
- Provided technical expertise on integrating new BI solutions and infrastructure monitoring tools (e.g., SPLUNK) to ensure the stability of data pipelines.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /Senior Data Engineer /MS BI Developer /EDW
Responsibilities:

- Designed and developed large-scale, complex enterprise-level software and Data Warehouse architecture, utilizing object-oriented programming (C++, C#, VB.NET) and T-SQL for core application and ETL development.
- Engineered robust ETL solutions for data extraction and transformation using advanced programming tools, loading complex, high-volume data into the data warehouse to serve business applications and reporting. (Generalizing SSIS/SSRS work).
- Designed and implemented scalable data warehouse architecture, including schema definitions, Fact and Dimension tables, and hardware infrastructure planning to optimize query performance.
- Developed and optimized SQL queries for data extraction, transformation, and reporting purposes, ensuring data accuracy and integrity across managed company databases.
- Built and deployed early machine learning models using techniques like k-nearest neighbors (k-NN) for customer churn prediction and fraud detection, achieving a 15% improvement in accuracy. (Highlighting relevant ML capability).
- Involved in the QA development lifecycle, including Unit Acceptance Testing (UAT) and system testing, by writing scripts (VBScript and batch files) to automate processes and ensure code reliability.
- Applied a systematic approach to data modeling and system architecture that is directly transferable to designing ETL pipelines for heterogeneous data types (Raster, Vector, Tabular) in modern cloud environments.

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Languages & Frameworks:** Java, JavaScript (including JQuery, jqGrid), C#, VB/VBA, AJAX, ASP.NET (Web Forms, MVC 2~6), ActionScript, XML, T-SQL, PL/SQL, SQL*Plus, ADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, .NET Framework/Core (2.0~6.0), AvePoint, K2 Black Pearl
- **OS**: Windows, Unix/Linux/BSD (Ubuntu 16~21, RHEL 3~10, SuSE 7~10, CentOS 5.0), Solaris 8~10, HP-UX 8~11, AIX 5.3/6.1
- **Databases & Data Platforms**: Oracle 9i/10gR2/11gR2/12c, Teradata, Exadata, Microsoft SQL Server 2005~2022, Azure SQL, Sybase, MySQL, Power BI
- **Cloud Platforms & Big Data**: Microsoft Azure, AWS Cloud (Amazon Athena, Redshift/Redshift Spectrum), Google Cloud Platform (GCP), Apache Spark, PySpark
- **Development & Data Tools / IDEs**: Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer, SQL Server Management Studio (SSMS) / BIDS, OBIEE 11g~12c, OWB 10gR2~11R2, Business Objects, Cognos

8~11, TM1 10.2, RStudio, R (3.5~4.3), Python (3.7~3.11), PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, Salesforce, MuleSoft Anypoint 7 and Mule 4.0

- **AI, Machine Learning, Deep Learning & BI Platforms**: Harrison Platform on Nvidia DGX-1 to DGX A100 systems, OpenAI API , ChatGPT-4o, Multi-GPTs, Auto-Agents, Amazon Bedrock (Claude/Titan), SageMaker & AWS Kiro; Azure OpenAI Service, Azure ML, Azure AI Studio, Copilot, Azure AI Foundry, Google Vertex AI, Gemini API (formerly MakerSuite) , Agent-2-Agent
- **Content Management Systems (CMS) & CRM**: SharePoint 2007~2022, CRM 3.0~D365, AX 2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco 4.x/6.x
- **Servers & Web/Application Servers**: Wintel Servers, Solaris Servers, Linux Servers, GlassFish, Apache Tomcat 6~10.1, WebSphere Application Server (WAS) / MQ, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization & Containerization**: Citrix Xen 5.0/5.5, VMware ESX 3.5~5/ESXi 5.5, IBM PowerVM, TM1,Hyper-V, Sun,Dell VIS
- **Security Technologies**: Microsoft Defender Suite, FortiGate Firewalls, FortiClient, Multi-Factor Authentication (MFA), SSL-VPN, IPSec VPN, Public Key Infrastructure (PKI), Digital Signing (RSA Data Security Inc.), Public Key Cryptographic Standards (PKCS#1), Hash functions (SHA-1, MD5)
- **Workflow & Automation Platforms**: K2 Workflow, Nintex 2010 Workflow, SharePoint 2007~2022 Workflows, AvePoint DocAve

Digital Transformation and AI Governance Lead – Parra- Gov- PERSOL – 5/Nov/2025
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Digital Transformation and AI Governance Lead /Software Engineer/Full Stack Developer(Java/.NET/Python) /Sr. DevOps Engineer /Snr Data Engineer/Mulesoft/AI/ML/DL/BI/SP/D365 Developer /EDW -Japan (7.5yrs) and Aust. (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Digital Transformation & AI Governance Lead/Software Engineer/Snr DevOps/Data Engineer/AI/ML Developer
Responsibilities:

- **I. Digital Transformation, AI Governance & Strategic Leadership**
- Spearheaded the development and execution of comprehensive digital transformation roadmaps, successfully integrating AI, automation, and Azure-based digital platforms to achieve key business goals such as reducing operational costs, improving customer experience, and accelerating time-to-market, resulting in enhanced service delivery.
- Drove the strategic planning and governance of AI initiatives by designing and overseeing Retrieval Augmented Generation (RAG) frameworks, ensuring large-scale knowledge bases are securely leveraged by Large Language Models (LLMs) while strictly adhering to NSW AI and data governance frameworks and ethical standards.
- Spearheaded the development of a next-generation digital ecosystem, engineering governed AI automation by deploying and managing agentic AI agents via the OpenAI Agent Builder, orchestrating complex workflows using n8n, and ensuring contextual trust by reliably consuming Model Context Protocol (MCP) services. E.g.: n8n trigger --webhook-id=mcp-agent-orchestration --data-payload '{"agent_id": "governance_agent_v2", "context_url": "https://mcp.api/v1/context/latest"}'
- Pioneered the use of Large Language Models (LLMs) and agentic workflows as optimizers within CI/CD pipelines to dynamically generate, test, and refactor code, resulting in a 25% reduction in deployment time and a significant improvement in code quality. E.g.,llm_output = bedrock_client.invoke_model(modelId=' anthropic.claude-4-sonnet-v1', or 'amazon.titan-text-express-v1' body={ "prompt": "Refactor this code to optimize performance: {code_snippet}"
- **II. Governance, Stakeholder Influence & Policy Support**
- Established and enforced robust governance frameworks, ensuring strict compliance with NSW AI & NSW data governance policies, ethical standards, assurance, and regulatory requirements across all emerging technology deployments.
- Policy & Data Management Support: Designed and led organisation-wide education and awareness programs to elevate capability and literacy in AI and digital governance, while supporting the design of crucial policies governing data quality, security, lifecycle management, and ethical usage.
- Project Oversight & Risk Management: Utilized strategic thinking and expert problem-solving abilities while collaborating extensively with legal, privacy, risk, and ICT stakeholders to secure the compliant and ethical deployments of emerging technologies. This involved leveraging excellent communication and negotiation skills to influence cross-functional teams and translate complex technical concepts into actionable strategies for non-technical stakeholders.

- Delivered Project coordination and assurance oversight for complex digital projects, including rigorous risk evaluation and monitoring of benefit realisation to ensure strategic investments drove measurable operational efficiency gains.
- Supported designing policies by providing technical assurance and governance oversight on data quality, security, lifecycle management, and ethical use for AI systems, specifically focusing on the performance and ethical considerations of data utilized within the LLM's Context Window.
- **III. AI/ML Engineering & Technical Implementation**
- Architected and developed a high-availability agentic AI core platform responsible for consuming and orchestrating Model Context Protocol (MCP) services, reducing inference latency by 15% across critical business workflows. E.g,:kubectl apply -f ai-agent-platform-deployment.yaml --context=az-aks-mcp-core
- Delivered highly scalable, fault-tolerant Azure systems, forming the secure MLOps backbone for Generative AI and enterprise automation workflows. E.g,: docker run -d --name ai-workflow -p 8080:80 my-genai-service:latest
- Expertly managed the integration of emerging AI technologies by utilizing LangChain and LangGraph to build robust, multi-step AI Agents and digital solutions, translating complex technical workflows into demonstrable enhancements in service delivery and operational efficiency.
- Developed, trained, and deployed a suite of machine learning and deep learning models on Amazon SageMaker using PyTorch, TensorFlow, and Hugging Face Transformers, automating model retraining to improve predictive accuracy and deliver NLP solutions.
- Built robust predictive models and secure AI APIs using diverse machine learning algorithms (e.g., KNN, CNN, regression) and ML paradigms (supervised, unsupervised) to drive Next Best Action (NBA) recommendations and pattern discovery.
- Developed and optimized data pipelines for AI model inference, enabling real-time predictions from large datasets, e.g., model.predict(new_data_batch).
- Proficient in modern development workflows, utilizing platforms like n8n for workflow automation, and MCP RAG for retrieval-augmented generation, alongside Git for version control and robust CI/CD practices for continuous integration and deployment. E.g.: n8n start (n8n), git push origin main (Git/CI/CD), mcps-rag run --model gpt4 (MCP RAG), and docker build -t myapp:latest . && docker push myapp:latest (CI/CD)
- Experienced developing, maintaining, and optimizing cloud-based applications ensuring high availability, scalability, and cost-effectiveness. aws_lambda_client.update_function_configuration(FunctionName='MyService', Memory=512)
- Developed, optimized, & secured applications, integrating zero-trust principles & security best practices to ensure reliability, scalability, and compliance. E.g: response.headers = 'max-age=31536000; includeSubDomains'
- Proficient in Infrastructure as Code (IaC), leveraging tools like Bicep or Terraform for efficient and consistent resource provisioning. E.g.: resource storage 'Microsoft.Storage/storageAccounts@2023-01-01' = { name: 'mystorageaccount' }

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Gov Central Bank: Digital Transformation & AI/ML Governance Lead /Snr DevOps/Data/Software Engineer/AI/ML Developer
Responsibilities:

- **High-Performance Financial Applications**: Designed & engineered high-performance, data-driven applications in Python (NumPy, SciPy, Pandas) to support complex financial operations and algorithmic strategy back-testing within the central bank's trading environment. Example: df['SMA_20'] = df['Close'].rolling(window=20).mean()
- **Advanced Statistical Modeling**: Developed and deployed predictive analytics and early machine learning models (utilizing Scikit-learn and initial TensorFlow/Keras implementations) for time-series forecasting, anomaly detection in trading data, and risk-assessment tasks. Example: model.fit(training_data, epochs=10, validation_split=0.2)
- **Enterprise API & Integration**: Acted as a core Full-Stack/Backend Engineer, designing and implementing robust RESTful APIs using Python (Flask/Django) and Java (Spring Boot) to facilitate seamless, real-time data exchange across the RBA's enterprise services layer, databases, and host systems. E.g,: POST /api/v1/orders HTTP/1.1 { "item": "widget", "qty": 2 }
- **SOA/Microservices Architecture**: Contributed to the strategic transition towards a Microservices architecture, focusing on building scalable, resilient, and cloud-agnostic application components to ensure high availability and business continuity. Example (Scaling): const cluster = require('cluster');
- **Full-Stack Development & Reporting**: Developed interactive, data-intensive Single-Page Applications (SPAs) using Angular/ TypeScript & JavaScript, ensuring robust integration with Python backends to deliver real-time monitoring dashboards & advanced analytical tools for regulatory compliance (e.g., APS standards). E.g.1: (Integration): axios.post('/api/report/data', payload) E.g. 2: (Compliance Logic): compliance_check_result = internal_ai_service.assess_trade_for_aps(trade_data)
- **DevOps & Automation**: Implemented CI/CD pipelines for application and model deployment, automating infrastructure provisioning and configuration management (CloudFormation/boto3) and integrating cloud messaging services for data flow consistency. E.g. 1: (IaC): template = cfn.get_template(StackName='my-iac-pipeline')
  E.g. 2: (Cloud Messaging): sqs_client.send_message(QueueUrl=queue_url, MessageBody=json_data)

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Digital Transformation & ML Governance Lead /Sr. DevOps/Data Engineer/Java/Mulesoft/D365 Developer
Responsibilities:

- **Machine Learning & Predictive Analytics**: Led the development of a data-driven system to predict international student enrollment likelihood. Engineered and deployed high-performance Machine Learning models using Python and R within an early Cloud-based platform (Azure ML Studio) to deliver core business forecasting.
  Example 1: predicted_students = predict_enrollment(prospective_students.copy())
  Example 2: model_output <- predict(trained_model, newdata)
- **Enterprise Integration Architecture(MuleSoft/ESB)**: Served as an Integration Solution Architect, designing and implementing a MuleSoft-based ESB framework to seamlessly integrate the modern student enrollment application with disparate legacy systems and third-party services, including orchestrating flows between Salesforce, SAP, and custom REST services for order processing.
- **Full-Stack Development & High-Volume APIs**: Built scalable, secure backend services using Node.js and developed performant applications using Python (Flask). Focused on robust RESTful API development and data security protocols for managing high volumes of concurrent applications. Example 1: const helmet = require('helmet'); app.use(helmet());
  Example 2: data = session.query(User).filter_by(status='active').all()
- **Modern UI Development**: Engineered dynamic, data-intensive Single-Page Applications (SPAs) for the student application portal using early React.js/Redux and advanced JavaScript, ensuring a responsive, intuitive user experience.
  Example: dispatch({ type: 'UPDATE_APPLICANT_STATUS', payload: 'submitted' })
- **DevOps & Cloud Orchestration**: Designed and maintained robust CI/CD pipelines using platform-specific tools to automate and streamline application deployment. Utilized Python and scripting for customized ETL/API orchestrations, ensuring high availability and system reliability. Example: aws s3 sync local_data/ s3://uc-student-data-bucket/ --delete

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: Python Application Developer /DevOps Engineer/Data Engineer /Software Test Engineer
Responsibilities:

- **Predictive Risk Modeling**: Developed and deployed advanced predictive models using Python (Scikit-learn) to accurately assess the likelihood of specific injury claim outcomes, fraud indicators, and resource allocation needs.
  Example 1: high_risk_claims = claims[model.predict_proba(features)[:, 1] > 0.7]
  Example 2: model_output = trained_model.predict(new_claim_features)
- **Data Engineering & ETL**: Drove core data initiatives by defining complex ETL requirements (including analyzing data sources, types, and formats) and creating data mapping documentation, implementing robust transformation logic to build the enterprise-wide data model for injury claim analytics and reporting.
- **Core Application Development**: Contributed to the development of the injury compensation claim system, designing and implementing features using Python and full-stack JavaScript (e.g., AngularJS/jQuery) to streamline user experience for claim management forms and dashboards. Example: app.post('/api/claims', async (req, res) => { /* secure claim intake */ });
- **Cloud Infrastructure Foundation**: Designed and developed foundational data solutions using early cloud services (AWS EC2, S3, RDS) to ensure the scalability and performance of the claim analytics platform within a hybrid environment.
- **Test Engineering & Multi-Language Expertise**: Ensured system quality and resilience by designing and executing comprehensive software test strategies. Maintained and debugged enterprise services, demonstrating expertise across languages including Java, Python, Go, C#, and PowerShell scripting for automation.
- Ensured system quality and resilience by designing and executing comprehensive software test strategies. Maintained and debugged enterprise services, demonstrating expertise across languages including Java, Python, Go, C#, and PowerShell scripting for automation.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Digital Transformation Lead DevOps Engineer/Data Engineer /Software Test Engineer/SP Consultant
Responsibilities:

- **Continuous Integration (CI) & Automation**: Designed and implemented early CI processes using Jenkins for build automation and deployment consistency. Developed extensive PowerShell and Python scripts to automate system provisioning, patching, and configuration management across virtualized infrastructure.
  Example 1 (Automation): subprocess.run(["powershell", "-file", "deploy_app.ps1"])
  Example 2 (Monitoring): aws cloudwatch get-metric-statistics --namespace AWS/EC2
- **Enterprise Data Governance (DBA/SharePoint)**: Served as a key data and systems specialist, responsible for database administration (DBA) and integrity. Managed the enterprise document and data repository within SharePoint, ensuring high availability, security, and compliance for critical council data. Example (Data Integrity): SELECT column_name, data_type FROM information_schema.columns WHERE table_name = 'council_data'
- **Software Quality & Test Engineering**: Led the testing lifecycle, designing and implementing large-scale logging, monitoring, and security measures to ensure application reliability. Used Java and Python for developing internal REST APIs and executing automated functional tests. Example (API Development): app.route('/api/v1/status', methods=['GET'])
  Example (Logging): logger.info("Application component initialized successfully.")
- **Infrastructure Management & Security**: Managed system infrastructure (physical and early cloud instances like AWS EC2 and S3) and implemented robust security best practices, including regular patching and application hardening to meet government security standards. Example (Storage/Data): os.path.join('data', 'processed', 'report.txt')

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Senior Systems Engineer /Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- **Automation & Scripted Delivery (Precursor to CI/CD)**: Led the improvement of continuous integration and delivery processes through advanced Shell and Python scripting, successfully reducing software feature delivery time and ensuring deployment consistency across environments. Example: bash deploy.sh --target=production --service=financial-reporting
- **Data Warehouse (DWH) Administration**: Functioned as a DBA, managing and monitoring the core Data Warehouse infrastructure for critical financial and insurance data. Ensured high availability, integrity, and performance necessary for regulatory reporting and business intelligence. Example: psql -c "VACUUM FULL ANALYZE;"
- **Systems Resilience & Security**: Served as the Senior Systems Engineer for Unix/Linux environments, responsible for the 24/7 operation and security hardening of servers hosting mission-critical applications in the finance domain.
  Example: cat /etc/sysconfig/iptables | grep 8080
- **Operational Efficiency**: Automated Tier 1 and Tier 2 support functions by designing and deploying the VMM Self-Service Portal, which optimized resource allocation and significantly minimized IT Helpdesk running costs.
  Example (Virtualization Management): Get-VM | Start-VM -TargetHost "HyperVHost01"
- **Enterprise Infrastructure Support**: Provided Level 2/3 technical support and administration for core Microsoft enterprise infrastructure, including Active Directory (AD), DNS, DHCP, Exchange 2010, and SharePoint.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /Unix/Linux Systems Administrator /DWH
Responsibilities:

- **Technology Transformation & Project Leadership**: Managed a critical initiative to replace legacy insurance applications (IBM AS/400 and LAMP stack) with a modern WISA-based platform. Strategic project was delivered to significantly reduce IT operational costs and enable integration with cloud-based Sales Force Automation (SFA).
  Example: SELECT platform_status, COUNT(*) FROM core_systems GROUP BY platform_status
- **Infrastructure Architecture & Systems Resilience**: Provided expert systems planning and implementation for all core IT infrastructure, including Hyper-V/VMware ESXi virtualization and high-availability SAN storage (HP/EMC/Dell).
  Example (Virtualization Management): Get-VMHost | Select-Object Name, ProcessorType, VmCount
- **UNIX/Linux Administration & Security Foundation**: Administered and monitored critical UNIX/Linux systems, ensuring the smooth, secure operation of applications supporting financial data processing. Implemented security principles and core software management best practices. Example (Resource Check): df -h /var/log/ | awk '{print $5}'
  Example (Security Configuration): grep "LISTEN" /etc/ssh/sshd_config
- **Data Warehouse (DWH) Support**: Acted as a key operational resource for the Data Warehouse environment, overseeing the core infrastructure that supported the company's vital business and financial reporting systems.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH
Responsibilities:

- **Large-Scale Data Warehouse (DWH) Architecture**: Led the planning, design, and development of complex, enterprise-level DWH solutions, including designing the hardware infrastructure landscape and defining detailed schemes, Fact and Relational Dimensions tables (Extended Star Schema), ensuring a robust architecture for global reporting requirements.
- **Enterprise ETL & Data Engineering**: Architected and built foundational, large-scale data pipelines (ETL using SSIS, T-SQL, C#, and VB.NET) essential for the organization's digital transformation towards advanced analytics. Pipelines processed complex raw data and loaded cleansed Master Data into Analysis Services Cubes (SSAS) for reporting.
- Established early enterprise standards for data quality assurance and cleansing, directly supporting today's requirements for governance, assurance, and reliable data lifecycle management (as required by current NSW AI and data policies).
- **Early Machine Learning (ML) Models**: Developed and deployed initial ML models using the k-nearest neighbors (k-NN) non-parametric supervised learning algorithm and Python for high-impact financial applications, including customer churn prediction and fraud detection.
- Business Intelligence (BI) & Reporting: Developed high-value BI solutions using the Microsoft stack (SSAS, SSRS). Managed reporting deployments and maintained underlying company databases, ensuring data accuracy and integrity for critical stakeholder requirements.
  Example (Querying Cubes): SELECT platform_status, COUNT(*) FROM core_systems GROUP BY platform_status

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Languages & Frameworks:** Java, JavaScript (including JQuery, jqGrid), C#, VB/VBA, AJAX, ASP.NET (Web Forms, MVC 2~6), ActionScript, XML, T-SQL, PL/SQL, SQL*Plus, ADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, .NET Framework/Core (2.0~6.0), AvePoint, K2 Black Pearl
- **OS**: Windows, Unix/Linux/BSD (Ubuntu 16~21, RHEL 3~10, SuSE 7~10, CentOS 5.0), Solaris 8~10, HP-UX 8~11, AIX 5.3/6.1
- **Databases & Data Platforms**: Oracle 9i/10gR2/11gR2/12c, Teradata, Exadata, Microsoft SQL Server 2005~2022, Azure SQL, Sybase, MySQL, Power BI
- **Cloud Platforms & Big Data**: Microsoft Azure, AWS Cloud (Amazon Athena, Redshift/Redshift Spectrum), Google Cloud Platform (GCP), Apache Spark, PySpark
- **Development & Data Tools / IDEs**: Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer, SQL Server Management Studio (SSMS) / BIDS, OBIEE 11g~12c, OWB 10gR2~11R2, Business Objects, Cognos 8~11, TM1 10.2, RStudio, R (3.5~4.3), Python (3.7~3.11), PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, Salesforce, MuleSoft Anypoint 7 and Mule 4.0
- **AI, Machine Learning, Deep Learning & BI Platforms**: Harrison Platform on Nvidia DGX-1 to DGX A100 systems, OpenAI (ChatGPT-4o, Multi-GPTs, Auto-Agents), Amazon Bedrock (Claude/Titan) & SageMaker, Azure OpenAI Service, Azure ML, Azure AI Studio, Copilot, Azure AI Foundry, Google Vertex AI, Gemini API (formerly MakerSuite) , Agent-2-Agent
- **Content Management Systems (CMS) & CRM**: SharePoint 2007~2022, CRM 3.0~D365, AX 2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco 4.x/6.x
- **Servers & Web/Application Servers**: Wintel Servers, Solaris Servers, Linux Servers, GlassFish, Apache Tomcat 6~10.1, WebSphere Application Server (WAS) / MQ, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization & Containerization**: Citrix Xen 5.0/5.5, VMware ESX 3.5~5/ESXi 5.5, IBM PowerVM, TM1,Hyper-V, Sun,Dell VIS
- **Security Technologies**: Microsoft Defender Suite, FortiGate Firewalls, FortiClient, Multi-Factor Authentication (MFA), SSL-VPN, IPSec VPN, Public Key Infrastructure (PKI), Digital Signing (RSA Data Security Inc.), Public Key Cryptographic Standards (PKCS#1), Hash functions (SHA-1, MD5)
- **Workflow & Automation Platforms**: K2 Workflow, Nintex 2010 Workflow, SharePoint 2007~2022 Workflows, AvePoint DocAve

Sr. AI Engineer– Bedrock-Hollard-TRC-26/10/2025
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Sr. AI Engineer– Bedrock-SageMaker Specialist / GenAI | AI Build | Automation Engineer/ Sr. DevOps /Data Engineer
Principle Software Engineer/Full Stack Developer(Py/Java/.Net/Mulesoft /AI/ML/DL/BI/SP/D365)-Japan (7.5yrs) & Aust. (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Sr. AI Engineer–Bedrock-SageMaker Specialist / GenAI | Principle Software Engineer | Automation Engineer
Responsibilities:

- **Professional Summary & Technical Expertise**
- Senior AI Engineer with 5+ years of experience in ML/Software Engineering, including 2+ years dedicated to architecting and deploying production-grade Generative AI and LLM solutions. Proven hands-on expertise in Python and the ML ecosystem, specializing in advanced RAG (including Graph RAG), prompt engineering, vector search, and model fine-tuning to drive business transformation.

- **Generative AI & Agentic Platform Development**
- Directed the end-to-end design, development, and high-availability deployment of production-grade Generative AI and LLM systems (including RAG and Graph RAG solutions) to automate key business use cases and drive operational transformation.
- Engineered and deployed mission-critical Python applications and GenAI platforms on a multi-cloud architecture (AWS, Azure, GCP), leveraging IaC, MCP+RAG, and n8n orchestration. E.g., refactored_code = json.loads(boto3.client('bedrock-runtime').invoke_model(modelId='anthropic.claude-4-sonnet-20240514-v1:0', body=json.dumps({"anthropic_version": "bedrock-2023-05-31", "max_tokens": 1024, "messages": [{"role": "user", "content": f"Refactor this code to optimize performance: {code_snippet}"}]}))['body'].read())['content'][0]['text']
- Engineered and productionized complex RAG and Agentic RAG pipelines, utilizing frameworks like LangGraph and LlamaIndex, integrating vector search (e.g., Pinecone/Weaviate) and Graph RAG to enhance multi-hop reasoning and achieve a 25% improvement in semantic query accuracy.

E.g.: text_splitter = RecursiveCharacterTextSplitter(chunk_size=1000, chunk_overlap=200).

- Developed and orchestrated sophisticated autonomous AI agents using frameworks like AutoGen and Semantic Kernel, and leveraging inter-agent communication protocols to integrate with external APIs, driving operational efficiency and intelligent decision-making. E.g., response = await kernel.invoke_async("email_plugin.summarize", text=email_body).
- **MLOps, Cloud Infrastructure & Backend Systems**
- Applied a builder's mindset to take full technical ownership of the LLMOps lifecycle, leveraging cloud platforms—specifically AWS (Bedrock, SageMaker)—to productionize scalable GenAI systems end-to-end, while automating CI/CD pipelines for 25% reduction in deployment time.
  E.g., boto3.client('lambda').invoke(FunctionName='AutomatedRemediationFunction', ...)
- Architected and deployed enterprise-grade Generative AI solutions on Amazon Bedrock, leveraging foundation models (e.g., Claude Sonnet 4, Amazon Titan) and integrating with core business systems via data-intensive APIs, reducing response time by 30%.
- Leveraged AWS Kiro's spec-driven development and agentic capabilities to design, build, and maintain scalable, secure AWS infrastructure (EC2, EKS, S3, RDS, Lambda, SageMaker, Amazon Neptune) for hosting advanced Graph RAG pipelines and LLMs.
- Designed & deployed scalable backend microservices using Node.js in TypeScript and Python frameworks (Django, Flask), implementing cloud-native architectures with AWS ECS/EKS to manage non-blocking I/O and facilitate seamless real-time AI inference.
- Designed & deployed scalable backend microservices using **Node.js** in **TypeScript**, enabling real-time data processing & seamless integration with AI-driven analytics pipelines built on platforms like LangChain, Amazon SageMaker, and Bedrock.
  - E.g.1: app.post('/stream-data', (req: Request, res: Response) => { req.pipe(new DataProcessorStream()).pipe(res); });
  - E.g.2: async function runAIPipeline(data: { source: string; }): Promise<{ prediction: string; confidence: number }> { return { prediction: 'anomaly_detected', confidence: 0.92 }; }
- **Strategic Leadership & Outcome Delivery**
- Collaborated cross-functionally with Product, Engineering, and Data teams to ensure AI efforts delivered tangible business outcomes (not just theoretical models), always prioritizing system maintainability, performance, and responsible AI use.
- Strategically framed complex Aged Care insurance business challenges—including claims automation and policy optimization—into defined ML tasks, emphasizing model interpretability and regulatory compliance in the final architecture selection.
- Drove the full lifecycle of AI solutions, from ideation and opportunity identification with stakeholders to the design and delivery of scalable prototypes, ensuring compliance with ethical AI standards.
- Consistently delivered high-quality AI solutions within agile sprint cycles, using data-driven metrics like MTTR and FPR to evaluate performance and communicate complex AI logic effectively to non-technical stakeholders.
- **Core Machine Learning & Automation**
- Developed scalable Python applications, leveraging Generative AI frameworks and speech-to-text models like Whisper to automate the transcription, summarization, and analysis of customer calls, optimizing deployment on AWS.
- Built robust predictive models and secure AI APIs using diverse machine learning algorithms (e.g., KNN, CNN, regression) and ML paradigms (supervised, unsupervised) to drive Next Best Action (NBA) recommendations and pattern discovery.
- Developed, trained, and deployed a suite of machine learning and deep learning models on Amazon SageMaker using PyTorch, TensorFlow, and Hugging Face Transformers, automating model retraining to improve predictive accuracy and deliver NLP solutions.
- Pioneered the use of Large Language Models (LLMs) and agentic workflows as optimizers within CI/CD pipelines to dynamically generate, test, and refactor code, resulting in a 25% reduction in deployment time and a significant improvement in code quality. E.g., llm_output = bedrock_client.invoke_model(modelId=' anthropic.claude-4-sonnet-v1', or 'amazon.titan-text-express-v1' body={ "prompt": "Refactor this code to optimize performance: {code_snippet}"

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Sr. AI/ML Engineer/Automation Engineer/Sr. DevOps/Data Engineer /AI/ML/DL/Mulesoft Developer
Responsibilities:

- **Core ML/DL & Advanced Automation Foundation**
- Pioneered Distributed AI: Developed self-learning Multi-Agent Systems (MAS), a subfield of distributed AI in Python, to automate and optimize complex algorithmic strategies for dynamic financial market pattern analysis.
- Engineered Intelligent Automation: Designed intelligent rule-based Robotic Process Automation (RPA) workflows, powered by core NLP libraries (NLTK, spaCy), to automate lead qualification and critical customer interaction analysis, enhancing data-driven decision-making.
- Developed scalable Python applications, leveraging frameworks like scikit-learn and NLTK for data indexing, feature extraction, & natural language processing. E.g., ml_model.predict(vectorizer.transform(["RBA rate change triggers volatility"]))
- **Deep Learning & Modeling**: Built robust predictive models and secure APIs using a diverse algorithm suite (KNN, CNN, RNN, GAN, supervised, unsupervised, reinforcement learning) applying Scikit-learn and Keras to drive risk mitigation and pattern discovery.

- Enhanced algorithmic trading strategies and risk mitigation by discovering new market principles beyond human intuition.Engineered scalable, mission-critical production applications in Python, driving robust system integrations and continuous deployment for machine learning services with core code such as requests.post(f"{api_url}/api/v1/deployments", json={'app':'ml-service', 'version':'v1.0.0'}).
- **Scalable Vision Systems**: Developed Computer Vision models using OpenCV to integrate image recognition for automated validation of property documents and anomaly detection, streamlining regulatory compliance workflows.
- **Audio Intelligence**: Led end-to-end ML solution development, integrating speech-to-text APIs for high-volume call transcription, automated summarization, and sentiment analysis within CRM workflows.
- **MLOps, Cloud-Native Deployment & Automation Excellence**
- MLOps & CI/CD: Engineered and productionized mission-critical applications in Python (Django, Flask), establishing scalable IaC CI/CD pipelines and isolated ML platforms to ensure robust deployment across AWS, Azure, and GCP.
- Cloud Architecture: Implemented a cloud-native architecture using AWS ECS for containerized microservices and REST APIs, enabling efficient model integration with enterprise platforms and streamlined data ingestion within a centralized data lakehouse.
- **Enterprise Automation**: Specialized in designing and implementing end-to-end automation solutions using Python & Red Hat Ansible Automation Platform (including Ansible Tower and AWX), developing modules and plugins to streamline secure, low-latency infrastructure provisioning for RBA trading platforms across hybrid cloud, on-prem, & RHEL9 environments. E.g., ansible-playbook deploy_low_latency_trading_engine.yml to automate trading engine deployments.
- **AIOps Integration**: Integrated PagerDuty with monitoring platforms (Datadog, Prometheus) to leverage AIOps/ML-powered operational insights, transforming alerts into actionable incidents and automating runbook remediation. E.g., aws cloudwatch put-metric-data --metric-name CriticalApplicationError --namespace MyAIOpsApplication --value 1 --unit Count --dimensions Environment=Production,Service=AuthService.
- **API Gateway Expertise**: Developed and deployed scalable applications using MuleSoft Anypoint Platform, focusing on robust backend API development and seamless system integrations for core banking systems.
- Developed and deployed scalable applications using MuleSoft Anypoint Platform, including Mule 4 & Anypoint Studio 7.x, with a strong focus on robust backend API development and seamless system integrations.
  E.g. 1: POST /api/v1/orders HTTP/1.1 { "item": "widget", "qty": 2 }
  E.g. 2: Flow orchestrates Salesforce, SAP, and custom REST services for order processing.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Sr. ML Engineer/Automation Engineer /Integration Solution Architect /ML/DL/Mulesoft Developer
Responsibilities:

- **Automated Systems and Machine Learning**
- Engineered self-learning Multi-agent systems (MAS) in Python to automate the management of student enrollment and financial data, enabling models to autonomously adapt to complex patterns for enhanced system performance.
- Engineered intelligent rule-based Robotic Process Automation (RPA) workflows powered by NLP libraries (e.g., NLTK, spaCy) for automated lead qualification and customer interaction analysis, improving data-driven decision-making.
- Developed computer vision models (using OpenCV and Scikit-image) to integrate image recognition for the automated analysis of visual data and anomaly detection, streamlining data processing and enhancing decision-making accuracy.
- Led end-to-end machine learning project development, from data collection and automated transcription to statistical analysis and intelligent summarization, ensuring efficient data flow for research and academic projects.
- Developed scalable Python applications, leveraging core frameworks like scikit-learn, NLTK, Theano, and Caffe for data preprocessing, modeling, and natural language processing. Utilized third-party speech-to-text services for automatic transcription and sentiment analysis, integrating results into custom-built data analysis pipelines.
- **Machine Learning and Data Insights**
- Built robust predictive models and secure APIs using diverse machine learning algorithms (KNN, CNN, RNN, ANN, regression, decision trees, random forests) and ML paradigms (supervised, unsupervised, reinforcement learning). Applied Scikit-learn and Keras for projects focused on pattern discovery and predictive analytics.
- Utilized GPU-accelerated systems to develop and optimize machine learning models for research projects. Developed and tested algorithmic strategies by applying emerging principles in statistical analysis.Integrated databases like PostgreSQL or MongoDB for managing and analyzing data in projects, enabling real-time and context-aware insights.
- **Systems Engineering and Automation**
- Implemented cloud-based architectures with AWS ECS for containerized microservices and REST APIs, ensuring efficient model integration and streamlined data ingestion.
- Automated the creation of secure and scalable infrastructure for machine learning experimentation using Infrastructure-as-Code (Terraform) and automated CI/CD pipelines (e.g., Jenkins) for robust deployment across cloud platforms like AWS, Azure, and GCP.
- Used AWS infrastructure (EC2, S3, and RDS) to design, build, and maintain scalable systems for hosting ML models.
- Designed and developed scalable backend APIs and integrations using Python frameworks (Django, Flask), optimizing data pipelines for efficient machine learning model training and enabling near real-time predictions on datasets.

- Developed an intelligent web scraper (Python, requests, BeautifulSoup, pandas) with a simple UI built using Django or Flask to demonstrate robust data acquisition and processing capabilities.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 and MS Dynamics 365 developer toolkit in VS 2015 to develop Dynamics CRM2013 applications, using customization & extension components to Streamlin data flows & integrate with SAP FinOps, S/4 HANA systems & using Liquid for templating and rendering dynamic content within web applications.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: ML Engineer /Automation Engineer/Software Test Engineer /DevOps/Data Engineer
Responsibilities:

- **Application Development and Machine Learning**
- Developed self-learning multi-agent systems in Python to automate and optimize the processing of injury claims, enabling autonomous adaptation to claims data and improving overall system efficiency.
- Developed and deployed predictive models using Python libraries like scikit-learn to accurately assess the likelihood of specific claim outcomes or potential fraud indicators, enabling proactive risk management and optimized resource allocation in the injury compensation domain.
- Python Application Developer with a robust quantitative foundation, adept at building high-performance solutions. Proficient in Python and Java, with capabilities to design, develop, and optimize ML applications and deliver actionable insights.
- Engineered secure and scalable backend services using Python (e.g., Flask) and Java to efficiently manage high volumes of injury compensation claim data, ensuring data integrity and seamless integration with critical EML systems.
- Designed and implemented user interfaces with jQuery and JavaScript, optimizing the claimant experience for EML's injury compensation portal and streamlining data presentation for complex claim workflows.
- Extensive expertise in programming languages such as Java, Python, C#, and PowerShell scripting languages.Data Engineering & ArchitectureArchitected and implemented data solutions on AWS, focusing on traditional data warehousing and ETL processes.
- Utilized Amazon EMR for batch processing of large data volumes, and AWS Redshift for near-real-time data analysis.
- Employed Amazon Kinesis as the unified service layer for real-time data ingestion and Amazon Redshift Spectrum for querying the data lake for advanced analytics and reporting.
- Designed and developed data solutions using AWS services like EC2, RDS, and S3.Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created and implemented mapping documents that defined how source data fields corresponded to target data fields, including writing code to map the data according to defined rules.
- DevOps & Automation
- Automated deployment of scalable machine learning applications and services, leveraging AWS services like EC2 to ensure high availability.Championed continuous integration and delivery (CI/CD) pipelines using tools like Jenkins for secure, continuous application delivery.
- Utilized PowerShell scripting to automate system configurations, testing, and deployment processes.Followed software development best practices for versioning, deployment, testing, and automation.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DevOps/Data Engineer /Data Engineer /Software Test Engineer/SharePoint Consultant /DBA
Responsibilities:

- Software Development and Deployment: Architected and built scalable web applications using Python and deployed them on traditional servers.
- Infrastructure Management: Managed infrastructure using scripting languages and automation tools.
- Application Development: Developed applications using Python for data processing and business logic.
- Data Management: Utilized relational databases for data storage and retrieval.
- System Integration: Implemented APIs and integrated with diverse systems.
- Automated build and deployment processes using Jenkins and shell scripting, managing server configurations with Puppet to ensure reliable delivery of machine learning models for research projects.
- Deployed and maintained web applications on virtual servers. Managed infrastructure using shell scripts and other automation tools, integrating them into CI/CD pipelines (e.g., Jenkins). Developed REST APIs using Python and Java.
- Monitored server resources using on-premise tools and implemented logging, monitoring, and security measures.
- **Security and Compliance**: Implemented security best practices and regularly patched systems and applications.
- **Agile and Data Expertise**: Experienced working in agile environments with strong data engineering practices.
- Other Proficiencies: Possessed a strong understanding of SQL, Python scripting, and SharePoint.

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Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Sr. Systems Engineer /Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- This role drove the modernization of core insurance systems through DevOps and automation, directly accelerating the delivery of new policy features, reducing underwriting risk, and improving operational efficiency across the policy lifecycle.
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- Key Responsibilities & Business Impact
- Accelerated Policy Feature Deployment (CI/CD): Spearheaded the implementation of Continuous Integration (CI) and automated deployment processes, resulting in a significant reduction in the time required to deliver new software features and regulatory updates for policy administration systems.
- Business Impact: Ensured the insurance products and customer portals could rapidly incorporate new underwriting rules and pricing models, maintaining market competitiveness.
- Infrastructure for Underwriting Scalability (IaC): Utilized scripting and configuration management tools to implement Infrastructure as Code (IaC) principles. This enhanced the scalability and reliability of the data warehouse (DWH) and core policy systems, which are crucial for timely risk analysis and claims processing.
- Reduced Operational Risk & Costs (Automation): Deployed the VMM Self-Service Portal to empower MLJ employees with self-service capabilities, minimizing IT Helpdesk costs while ensuring critical DevOps and Unix/Linux systems supporting policy data were stable and available.
- System Security for Compliance: Ensured robust security measures were in place across the expanding infrastructure, safeguarding sensitive customer and policy data to maintain strict regulatory and data compliance standards.
- Business Continuity and Support: Administered and monitored the core Unix/Linux-based systems and provided Level 2/3 support for the entire Microsoft infrastructure (AD, Exchange, SharePoint), ensuring the smooth, continuous operation of all critical business and policy-related applications.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /Unix/Linux Systems Administrator
Responsibilities:

- This role combined technical project leadership with deep operational support, directly impacting the underwriting and policy administration processes through critical systems upgrades, driving business continuity and process optimization.
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- Key Responsibilities & Business Impact
- Underwriting and Policy System Modernization: Managed the end-to-end project to replace the legacy IBM AS/400-based insurance web application with a modern WISA-based system, a project critical for improving the efficiency and compliance of new policy underwriting and customer data management.
- Sales Process Integration & Analysis: Successfully integrated the new insurance platform with the Salesforce Automation (SFA) platform, enabling real-time data flow for sales performance analysis and providing underwriters with timely, accurate application data, which was essential for risk assessment.
- Infrastructure for Business Continuity: Planned, implemented, and administered core IT systems (Hyper-V, VMware, SAN Storage, Windows Server/Linux) that maintained the 24/7 high-availability of the Data Warehouse (DWH) and underwriting applications. This ensured business continuity for policy processing and regulatory reporting.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Sr. Systems Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for Business Intelligence (BI) solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized C++, C#, VB.NET, and T-SQL for SSIS and SSRS programming.
- Designed data warehouse architecture, including hardware infrastructure, schema definitions, and Fact and Relational Dimensions tables.
- Developed data warehouse and business reporting solutions to meet stakeholder requirements using SSIS, SSAS, and SSRS.
- Used SSIS ETL tools to extract and transform complex data, loading it into the data warehouse to produce Analysis Services Cubes for business reporting via SSRS or Data Marts to serve business applications.
- Created and deployed reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio, including debugging, monitoring, and troubleshooting BI solutions.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Managed and maintained company databases, ensuring data accuracy and integrity.
- Defined the target warehouse module, and created target user, module, and schema.
- Built and deployed machine learning models using k-nearest neighbors (k-NN) for customer churn prediction and fraud detection, achieving a 15% improvement in accuracy.

- Involved in UAT and TEST in a QA development environment by writing VBScript and batch files to automate processes.

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Languages & Frameworks:** Java, JavaScript (including JQuery, jqGrid), C#, VB/VBA, AJAX, ASP.NET (Web Forms, MVC 2~6), ActionScript, XML, T-SQL, PL/SQL, SQL*Plus, ADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, .NET Framework/Core (2.0~6.0), AvePoint, K2 Black Pearl
- **OS**: Windows, Unix/Linux/BSD (Ubuntu 16~21, RHEL 3~10, SuSE 7~10, CentOS 5.0), Solaris 8~10, HP-UX 8~11, AIX 5.3/6.1
- **Databases & Data Platforms**: Oracle 9i/10gR2/11gR2/12c, Teradata, Exadata, Microsoft SQL Server 2005~2022, Azure SQL, Sybase, MySQL, Power BI
- **Cloud Platforms & Big Data**: Microsoft Azure, AWS Cloud (Amazon Athena, Redshift/Redshift Spectrum), Google Cloud Platform (GCP), Apache Spark, PySpark
- **Development & Data Tools / IDEs**: Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer, SQL Server Management Studio (SSMS) / BIDS, OBIEE 11g~12c, OWB 10gR2~11R2, Business Objects, Cognos 8~11, TM1 10.2, RStudio, R (3.5~4.3), Python (3.7~3.11), PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, Salesforce, MuleSoft Anypoint 7 and Mule 4.0
- **AI, Machine Learning, Deep Learning & BI Platforms**: Harrison Platform on Nvidia DGX-1 to DGX A100 systems, OpenAI (ChatGPT-4o, Multi-GPTs, Auto-Agents), Amazon Bedrock (Claude/Titan) & SageMaker, Azure OpenAI Service, Azure ML, Azure AI Studio, Copilot, Azure AI Foundry, Google Vertex AI, Gemini API (formerly MakerSuite) , Agent-2-Agent
- **Content Management Systems (CMS) & CRM**: SharePoint 2007~2022, CRM 3.0~D365, AX 2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco 4.x/6.x
- **Servers & Web/Application Servers**: Wintel Servers, Solaris Servers, Linux Servers, GlassFish, Apache Tomcat 6~10.1, WebSphere Application Server (WAS) / MQ, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization & Containerization**: Citrix Xen 5.0/5.5, VMware ESX 3.5~5/ESXi 5.5, IBM PowerVM, TM1,Hyper-V, Sun,Dell VIS
- **Security Technologies**: Microsoft Defender Suite, FortiGate Firewalls, FortiClient, Multi-Factor Authentication (MFA), SSL-VPN, IPSec VPN, Public Key Infrastructure (PKI), Digital Signing (RSA Data Security Inc.), Public Key Cryptographic Standards (PKCS#1), Hash functions (SHA-1, MD5)
- **Workflow & Automation Platforms**: K2 Workflow, Nintex 2010 Workflow, SharePoint 2007~2022 Workflows, AvePoint DocAve

AI Engineer – Cybersecurity-EKS-Avertro – 16/10/2025
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior AI Engineer / GenAI | Automation Engineer/ Software Engineer /Sr. DevOps /Data Engineer /Full Stack Developer (Python/React/Node.js/GraphQL /Java/.Net/Mulesoft /AI/ML/DL/BI/SharePoint/Dynamics 365)-Japan (7.5yrs) & Aust. (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present | Aged Care Quality Safety and Commission (ACQSC), Sydney
Gov. Health: Sr. AI Engineer | Automation Engineer | Software Engineer | Full Stack Developer | Sr. DevOps /Data Engineer
Responsibilities:

- **AI Vision, Strategy & Product Leadership**
- Spearheaded the design, build, and scaling of Generative AI and agentic LLM-powered solutions across multiple cloud platforms, driving the full lifecycle of AI solutions and delivering cyber risk and operational intelligence.
- Transformed raw security and proprietary business data into clear, actionable intelligence using Python, spaCy, and Hugging Face Transformers, directly supporting executive decision-makers in quantifying and managing cyber risk.
- Functioned at the intersection of NLP, data enrichment, and product design to implement robust text processing pipelines, ensuring seamless integration of intelligent features into core software products.
- **Generative AI, RAG & Agentic Workflows**
- Engineered & optimized high-impact LLM-driven solutions for cyber risk intelligence, including multi-document summarization, structured data extraction, and autonomous workflow automation.
- Architected and scaled a production-grade RAG framework featuring advanced context management and layered prompt engineering, achieving high-fidelity risk posture data retrieval. Integrated an LLM-as-an-Optimizer module to automatically test and refactor retrieval strategies based on real-time performance metrics, dramatically improving remediation efficacy and response quality. E.g., llm_output = bedrock_client.invoke_model(modelId='anthropic.claude-v4', ...)
- Architected and deployed low-latency, production-grade Retrieval-Augmented Generation (RAG) pipelines, leveraging specialized tooling (e.g., LangChain, LlamaIndex, vector databases) with optimized chunking strategies & asynchronous endpoints (FastAPI) for real-time performance monitoring (Langfuse). Example: response = RAGChain.run({"question": security_vulnerability, "docs": retrieved_context})
- Pioneered and optimized high-performance, end-to-end RAG pipelines, deploying LangChain for orchestration and intelligent data chunking (E.g. RecursiveCharacterTextSplitter(chunk_size=1000, chunk_overlap=200)). Specialized in advanced retrieval architectures, including Knowledge Graph (KG) RAG, to enhance contextual understanding and answer complex semantic queries (E.g. async def query_llm_with_rag(index: VectorStoreIndex, question: str) -> Dict[str, Any]:), resulting in a 20% reduction in API calls and boosting query performance.
- Developed and orchestrated sophisticated autonomous AI agents using frameworks like LangGraph, AutoGen, and Semantic Kernel, and leveraging inter-agent communication protocols and tools like n8n to integrate with external APIs, driving operational efficiency and intelligent decision-making. E.g., response = await kernel.invoke_async("email_plugin.summarize", text=email_body).
- Achieved end-to-end cyber risk autonomy by architecting a coordinated system of Agentic AI Agents that translate qualitative Customer Needs into quantified, financial risk exposure using a proprietary Cyber Risk Quantification (CRQ) Engine.
- Pioneered proactive cyber defense by integrating Attack Graph Simulators with the core Agentic AI Agent to automatically test and optimize multi-phased mitigation strategies, ensuring all Tailored Proactive Solutions are financially justified.
- Automated solution design and deployment across the GRC lifecycle, leveraging Data Acquisition Agents to pull real-time SAFE Security Data and enabling the primary Agentic AI Agent to autonomously generate an actionable remediation Roadmap directly within the SOAR/GRC Platform.
- Architected and implemented a serverless MLOps workflow for a core cybersecurity anomaly detection agent, leveraging AWS EventBridge and Lambda to establish a fully autonomous, event-driven operational loop.
- Designed the EventBridge Trigger system to execute the Monitoring Agent on a scheduled basis (hourly/daily), enabling continuous, programmatic validation of model performance and drift.
- Ensured sustained detection efficacy by automating performance-based retraining triggers directly from the monitoring results, drastically reducing the time required to adapt the detection agent to evolving threats.
- **Core ML & Data Pipeline Engineering**
- Designed and implemented robust, end-to-end text data processing pipelines and ML workflows to analyze, classify, and enrich vast volumes of unstructured data.
- Applied advanced NLP techniques and pattern recognition models to extract key insights, identify critical trends in textual data, and generate meaningful intelligence for enterprise-level decision support.
- Developed and trained robust machine learning models (prediction, anomaly detection, and regression) for high-impact cybersecurity applications, leveraging deep feature engineering expertise across structured and unstructured datasets. E.g.1: clf.fit(security_log_features, incident_classification_labels)
- Developed and operationalized real-world AI applications focused on high-volume unstructured text, leveraging spaCy for fast, production-scale Named Entity Recognition (NER) and Hugging Face for generating numerical embeddings and fine-tuning state-of-the-art transformer models.
  Example: embeddings = model(**tokenized_input)['pooler_output'].detach().cpu().numpy()
- Architected and managed the full-stack infrastructure for a Cyber Risk Management Platform (React/Node.js/GraphQL), developing responsive dashboards that leverage Apollo client hooks to consume structured data models, delivering real-time threat posture (e.g., Prompt Injection, Phishing) to the UI. Key features included dynamic Role-Based Access Control (RBAC), financial risk modeling, and integrated remediation workflow automation across scaled AWS services (ECS, Lambda, and DynamoDB). Example: const { data } = useQuery(GET_RISK_POSTURE);
- **MLOps, Deployment & Cloud Architecture**
- Architected & implemented production-grade monitoring & evaluation (M&E) frameworks for high-volume LLMs & predictive ML models, leveraging specialized observability tooling (e.g., Langfuse) & cloud-native services (e.g., Amazon CloudWatch)

to track performance drift & enforce data quality and security guardrails across scalable deployment pipelines. E.g., Langfuse.trace_guardrails(response, checks=["PII_redaction", "hallucination_score"])

- Integrated ML/AI services into core applications using modern cloud and containerized deployment methods (e.g., Docker, AWS ECS/EKS/Lambda), ensuring high availability and robust scalability. Leveraged Kubernetes for complex orchestration (e.g., kubectl apply -f k8s/rag-pipeline.yaml).
- Pioneered the use of LLMs and agentic workflows to automate and refactor code within scalable CI/CD pipelines, reducing deployment time by 25%. Engineered and deployed mission-critical Python applications and GenAI platforms on a multi-cloud architecture (AWS, Azure, GCP), leveraging IaC, MCP+RAG, and n8n orchestration. E.g.1: llm_output = gemini.generate_text(f"Refactor this code to optimize performance: {code_snippet}")
- Leveraged AWS Kiro's spec-driven development & agentic capabilities to design, build, and maintain scalable, secure AWS infrastructure (EC2, EKS, S3, RDS, Lambda, SageMaker, Amazon Neptune) for hosting advanced Graph RAG pipelines and vector databases.

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Gov Central Bank: Senior AI/ML Engineer & DevOps Lead (Government Central Bank)
Responsibilities:

- **Technical Foundations & Advanced Modeling**
- Pioneered the Bank's predictive modeling capabilities, building and securing deep learning architectures (CNN, RNN, GAN, Transformer models) and classical ML pipelines using PyTorch, TensorFlow, and Keras to drive pattern discovery and autonomous decision-making in sensitive environments.
- Applied advanced NLP and computer vision techniques (NLTK, spaCy, OpenCV) for high-volume text analysis, intelligent summarization, and anomaly detection in financial data flows.
- Engineered scalable, production-grade ML APIs in Python (Django, Flask), establishing the foundation for robust, low-latency system integrations and continuous model deployment.
- Led the end-to-end development of data pipelines, from secure audio ingestion and automated transcription to intelligent summarization, ensuring secure and seamless data flow across multi-cloud environments.
- Architected and managed the full-stack infrastructure for a Cyber Risk Management Platform (React/Node.js), developing responsive risk dashboards that utilized AJAX/Fetch calls to consume RESTful API endpoints from a high-performance Express.js/Node.js layer, delivering real-time threat posture (e.g., SQL Injection, Phishing) to the UI. Key features included dynamic Role-Based Access Control (RBAC), configuration of financial risk modeling parameters, and integrated remediation workflow automation across scalable AWS services (e.g., EC2, S3, and RDS).
- **MLOps, Deployment & Infrastructure at Scale**
- Automated secure, scalable Infrastructure-as-Code (IaC) CI/CD pipelines, enabling robust multi-cloud deployment (AWS, Azure, GCP) for server-side ML applications.
- Implemented cloud-native architectures utilizing AWS ECS for containerized microservices and REST APIs, ensuring efficient model serving and streamlined data ingestion within a Data Lakehouse architecture (S3, Redshift).
- Specialized in designing & implementing full-stack automation solutions using Python and Ansible to streamline the secure, low-latency provisioning of core production systems. (e.g., ansible-playbook deploy_low_latency_trading_engine.yml)

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior ML Engineer & Security Automation Architect
Responsibilities:

- **Enterprise Security & AI-Driven Anomaly Detection**
- Architected and Deployed a Multi-Agent System (MAS) written in Python to autonomously monitor high-volume student enrollment and finance records, flagging high-risk anomalies (e.g., bulk enrolments, payment deviations) indicative of account compromise and fraud.
- Developed advanced computer vision models (OpenCV, Scikit-image) for non-textual anomaly detection, applied to identifying complex security breaches and behavioral patterns within large, structured data logs.
- Built secure predictive models and enterprise APIs (Django/Flask) utilizing JWT for granular access control, applying diverse ML algorithms (CNN, RNN, ANN, regression) to secure and manage sensitive student data.
- **MLOps and Cloud Systems Integration**
- Pioneered the use of Infrastructure-as-Code (IaC) with Terraform to automate the creation of secure, isolated infrastructure & CI/CD pipelines (Jenkins) for ML experimentation & robust deployment across multi-cloud platforms (AWS, Azure, GCP).
- Engineered intelligent rule-based automation (RPA) workflows powered by Python and NLP libraries (NLTK) to automate complex data analysis, preprocessing, and model training tasks at scale.